

Projects

Bun2nix | Rust | Nix | JavaScript

Open Source Tool for Building Bun Projects with Nix

- Developed **40x faster** alternative to existing JS packaging solutions by leveraging Rust's performance combined with Bun's
- Reverse-engineered Bun's package resolution logic to create reproducible Nix expressions
- Integrated with Bun's native APIs to enable real-time dependency tracking
- Contributed to lang2nix ecosystem with **100+ people** checking out the project

Hack Ireland Winner | Intercom | Ruby on Rails | AI

€2000 Human Computer Interaction Prize (Feb 2025)

- Led development of **AI-powered** knowledge management system using multi-modal input (voice/text)
- Implemented reproducible shared environment using **devenv** for environment consistency
- Had **35/64** commits and won a prize despite two of my teammates leaving partway through
- Live packaged text to speech software from source in **Nix** for orchestration with the project

NixOS Operating System Configuration | Linux | DevOps

Reproducible System Configuration

- Implemented **atomic rollbacks** for critical services using NixOS module system
- Automated updates for **10+ packages** and provided documentation with CI/CD pipeline integration
- Implemented wrappers around various utilities to make my config portable across multiple systems

Rust based Web-App | Leptos | Actix

High-Performance Web Application

- Achieved **99.9% type safety** across 5k+ LOC Rust codebase
- Designed **<100ms** page loads through WASM-optimised Leptos components
- Implemented **zero-cost API abstractions** making deployment costs very cheap and easy to scale
- Reduced deployment artefacts by **90%** using Nix over our original Docker image

Work Experience

MERN Full Stack Web Developer

Bar Monkey Calisthenics: Jan 24 > July 24

- Shipped **4 major features** including timetables, customer progress tracking and payment integration
- Achieved **98% Lighthouse performance score** through **Preact** optimization
- Containerised deployment pipeline allowed for **close to 0 runtime cost** on **fly.io** for the user base

Data Management Intern

AIB: June 23 > August 23

- Digitised **200+ historical records** with **99.8% accuracy** rate
- Learned crucial skills for interacting with data in an office setting
- Implemented Microsoft Word macros accelerating letter generation by **5x**

Skills

- **Languages:** Rust (Advanced), JavaScript/TypeScript (Advanced), SQL (Intermediate) , Python (Intermediate), Bash (Intermediate)
- **DevOps:** Nix/NixOS, Docker, GitHub Actions, AWS
- **Web:** React, Node.js, Leptos, Actix, MongoDB
- **Concepts:** Type Systems, Functional Programming, Performance Optimisation

Education

Institution	Goal	Dates Attended	Outcome
Trinity College Dublin	B.A. Computer Science	Sep 23 > May 27	81% Grade Average
Clonkeen College	Leaving Cert	Sept 18 > June 23	601 Points